

Computer Vision Project Milestone

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Vision-Language Models for E-Commerce Product
Description Generation

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1. Problem Context

The task addressed in this project falls under computer vision-based image classification combined with vision-language generation, where the goal is to automatically generate meaningful product descriptions from images and associated metadata. In modern e-commerce platforms, such as online marketplaces, product descriptions play a crucial role in influencing customer decisions by clearly communicating features, materials, and usage. However, manually writing these descriptions is labor-intensive, time-consuming, and difficult to scale as product catalogs grow rapidly.

From a computer vision perspective, the core challenge is not just recognizing objects in an image but understanding fine-grained product attributes, such as texture, style, or material. Unlike traditional image captioning tasks that generate neutral, factual descriptions, product descriptions must be informative, structured, and persuasive, often combining both visual cues and non-visual metadata. Existing models struggle because they primarily rely on visual input and fail to incorporate structured product details like brand, dimensions, or specifications effectively.

Additionally, real-world challenges include domain variability (different product categories), limited labeled datasets for local markets like Pakistan, and the need for context-aware language generation. Therefore, building a system that integrates computer vision classification with NLP-based description generation is highly relevant and remains an open research problem in multimodal AI.

2. Research Review

2.1 CLIP (Learning Transferable Visual Models From Natural Language Supervision) - NeurIPS 2021

CLIP aimed at learning image representations directly from raw text. This enables zero shot transfer to downstream tasks without dataset specific training. CLIP trains an image encoder, either ResNet or Vision Transformer) along with text encoder, Transformer, using a contrastive image. It maximises the cosine similarity of correct image text pairs all while it minimizes the incorrect pairings. The dataset used was the WIT (webimage text). This dataset is a newly constructed dataset containing 400 million image to text

pairs collected from the internet. CLIP matches the accuracy of the original ResNet-50 on imagenet zero-shot CLIP(76.2%) without using any of the 1.28 million training examples [1]. The model struggles with complex or abstract tasks, such as satellite image classification, counting objects, and fine-grained classification like traffic sign recognition. It also generalizes poorly to out-of-distribution data, such as the handwritten digits in the MNIST dataset.

2.2 Vision Transformer (ViT) - ICLR 2021

The reliance on convolutional neural networks (CNN) is unnecessary and a pure transformer applied directly to images patches can perform exceptionally better when image classification is the main goal, is what the authors deduce. The approach splits an image into fixed size patches and provides the sequence of linear embeddings of these patches as input to a standard transformer encoder. Image patches are treated as tokens are treated in natural language processing. Dataset used: ImageNet and ImageNet-21k. As well as an in-house JFT-300M dataset. When pre-trained at a sufficient scale, the best ViT model reached an accuracy of 88.55% on ImageNet, 94.55% on CIFAR-100, and 77.63% on the VTAB suite [2]. Great results, low computation. When trained on mid sized datasets without strong regularization, ViT yields modest accuracies although below ResNets of similar sizes.

2.3 BLIP (Bootstrapping Language Image Pre-training) 2022

A unified vision language pre training framework that transfers flexibly to both understanding and generation based tasks. The architecture used is a Multimodal Mixture of Encoder-Decoder (MED) that operates as a unimodal encoder. It utilizes a dataset bootstrapping method called CapFilt. CapFilt ensures a captioner generates synthetic captions and a filter removes noisy ones. Dataset used: Pre training utilized 14 million images from datasets like COCO, visual Genome, Conceptual captions, and SBU. As well as, the huge 115 million images LAION400M. BLIP achieves state of the art results across various tasks showing a 2.7% increase in average recall@1 for image text retrieval on COOC and outperforming ALBEF by 1.6% on the VQA test set [3]. It also demonstrates strong zero shot transfers, transfer capabilities for video language tasks like text to video retrieval. The biggest limitation is the parameter sharing between the caption and filter during the cap filled process will be the confirmation bias which decreases his performance on dawn stream tasks, the quality of synthetic captions relies heavily on diversity, making stochastic nuclear sampling far superior to deterministic beam search.

2.4 BLIP-2 2023

Blip2 provides the generic and efficient strategy that bootstraps vision, language pre-training from off the shelf models, the frame work bridges, the modality gap using a

lightweight querying transformer. The pre-training is divided into two stages, bootstrapping vision language representation learning from frozen image encoder and bootstrapping vision to language generative learning from a frozen LLM. The pre-training data set consists of 129 million images incorporating sources like COCO, visual genome, CC3M, and CC 12, SBU LAION400M. the model achieves incredible results on various vision language tasks such as the model achieves the model achieve model achieve the model the model. The the model achieves great results on various vision, language tasks with significantly fewer trainable parameters and existing methods specially blip2 out performs Flamingo80B by 8.7% on zero short VQAV2 while utilizing 54X fewer trainable parameters context learning [4] via a few short examples does not improve VQA performance because the pre-training data set only contains a single image text pair per sequence. It also suffers from frozen LLMs, such as hallucinating, incorrect knowledge or propagating, social bias.

2.5 LLaVA (Large Language and Vision Assistant) 2023

LLaVa aims to extend the instruction tuning to the multimodal space by connecting a vision encoder and an LLM to build a general purpose visual plus language assistant. The authors use language only GPT-4 to generate multimodal language image instruction following data. The architecture Connect the open, set visual encoder of CLIP with the language decoder Vecuna using a simple linear layer and the entire model is fine tuned and dataset used is 595K filtered image text pairs from CC3M for feature alignment and 158K unique language image instruction following samples using COCO images. LLaVA yields an 85.1% relative score compared with GPT-4 on a synthetic multimodal instruction-following dataset. When ensembled with GPT-4 and fine-tuned on Science QA, the model achieves a new state-of-the-art accuracy of 92.53% [5]. LLaVa sometimes perceived an image as a bag of patches and failed to grasp complex semantics. An example would be the hallucination centering a specific flavor of yogurt just because nothing yogurt and the flavoring fruit are present in the image. It also inherently retains the basis and hallucination risks passed down from its base models

2.6 InstructBLIP 2023

The core objective of this work was to develop a general-purpose vision-language model (VLM) capable of handling a wide range of tasks using natural language instructions. It addresses key limitations of existing models, which struggle with diverse visual inputs and lack strong instruction-following capabilities in multimodal settings. The architecture is based on the BLIP-2 framework and consists of three main components: a frozen image encoder, a frozen large language model (LLM), and a trainable Query Transformer (Q-Former). The key innovation is the Instruction-Aware Q-Former, which takes both the image and the instruction as input and extracts task-relevant visual features dynamically. The model is trained on 26 datasets spanning 11 task categories, including image caption-

ing (COCO), visual question answering (VQAv2), and visual reasoning (GQA) among others. In terms of performance, the key result is its strong zero-shot capability, achieving state-of-the-art (SOTA) results on all 13 held-out datasets and outperforming prior models such as BLIP-2 and Flamingo, with approximately a 15% relative improvement over BLIP-2. The evaluation metrics used include Accuracy, CIDEr, and AUC. Additionally, when fine-tuned, the model achieves SOTA performance on several benchmarks, such as ScienceQA (90.7% accuracy) [6]. However, the model has certain limitations. It inherits common weaknesses of LLMs, such as hallucinations and biased outputs, and requires careful evaluation for fairness and safety before deployment. Furthermore, the training process is complex, involving multiple datasets, careful balancing strategies, and significant computational resources. Overall, InstructBLIP represents a significant step toward building general-purpose multimodal AI systems.

2.7 ALIGN (Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision) 2021

The objective of this work was to develop a vision-language model (VLM) capable of advanced multimodal understanding (image + text). Specifically, it aimed to align visual features with a powerful large language model (LLM) to enable the generation of detailed image descriptions and support visual reasoning. The base architecture combines a vision encoder from BLIP-2, a large language model (Vicuna), and a trainable linear projection layer. Instead of training the entire model, only this projection layer is trained, which maps visual features into the language embedding space and acts as a bridge between image and text. The training process is divided into two stages. In Stage 1, large-scale image-caption datasets such as Conceptual Captions and LAION are used; these datasets are noisy but highly diverse and help achieve initial alignment. In Stage 2, a high-quality curated dataset of approximately 3.5K samples is used, which is manually refined to improve response quality and conversational ability [7]. In terms of performance, the model demonstrates the ability to generate rich and detailed image descriptions, along with strong conversational capabilities. It also shows improved language fluency compared to earlier VLMs and outperforms models like BLIP-2 specifically in conversational quality. The evaluation is primarily qualitative, relying on human judgment rather than standard benchmark metrics. However, the model has several limitations. It suffers from hallucinations, sometimes generating incorrect details about images. Additionally, it lacks strong quantitative evaluation, depends heavily on the underlying LLM, and is constrained by the relatively small dataset size in Stage 2. Overall, ALIGN demonstrates that combining a simple projection-based alignment mechanism with a strong LLM can result in a powerful and cost-efficient vision-language model capable of high-quality multimodal conversation.

2.8 Flamingo 2022

The primary objective in building Flamingo was to develop a Visual Language Model that can rapidly adapt to novel multimodal tasks using few shot learning with only a few annotated examples. To achieve this, Flamingo leverages a unified architecture that bridges powerful, pretrained, frozen vision only models (NFNet-F6) and language only models (Chinchilla) to preserve the extensive knowledge gained during their respective pretraining phases. The model utilizes a Perceiver Resampler to map a variable number of visual features into a fixed set of 64 visual tokens, significantly reducing computational complexity. Furthermore, the integration of gated cross attention layers and multi visual masking ensures architectural stability and allows the model to handle sequences of arbitrarily interleaved visual and textual inputs. Training was conducted on the M3W (MultiModal MassiveWeb) dataset, which contains interleaved text and images, alongside paired image text datasets such as LTIP and LAION. Results showed that with only 32 examples, Flamingo set a new few shot state of the art on 16 multimodal benchmarks; notably, the 80B Flamingo model outperformed the best available fine tuned models on six of these tasks despite using approximately 1,000 times less task specific data [8]. However, the model faces limitations such as catastrophic forgetting where unfreezing the language model components causes it to lose its original pretraining knowledge and the high computational intensity required for inference and training.

2.9 OFA 2022 - ICML

The objective of OFA was to create a unified multimodal paradigm that eliminates the need for complex, task specific or modality specific customizations, enabling a single system to handle a variety of tasks such as classification, generation, and self supervised pretext tasks. The methodology followed a sequence to sequence framework using hand-crafted instructions and a unified vocabulary for all linguistic and visual information, including subwords, image codes, and location tokens. The model utilized an encoder decoder transformer backbone supported by advanced positional encoding. For pretraining, the image text paired datasets used included Conceptual Captions (CC12M), SBU, and Visual Genome, while vision only data consisted of raw images from OpenImages and ImageNet 21K for image infilling and object labelled data from Object365 for detection tasks. Key metric results for cross modal tasks included a score of 82 on VQA and 91.2 on the SNLI VE test; furthermore, the model achieved a CIDEr score of 154.9 for image captioning on the MSCOCO dataset [9]. Some limitations were noted in sentence pair classifications, where the model did not achieve satisfactory performance on zero shot tasks (accuracy under 60 percent) likely due to a lack of relevant pretraining data. Additionally, model performance proved highly sensitive to the design of manual instructions, as slight changes could drastically and inconsistently influence results. Overall, OFA provides a unified framework for multimodal pretraining that aims to satisfy the three properties of being task agnostic, modality agnostic, and task comprehensive.

2.10 ECLIP (E-Commerce Contrastive Language-Image Pre-training) 2023 - CVPR

The objective of this work was to establish a generic foundation model for e-commerce that can scale to a wide range of downstream applications like product retrieval and item classification. It aimed to overcome the limitations of general domain models like CLIP when applied to e-commerce, where the correlation between text and images is often much sparser. The methodology employed an encoder decoder architecture consisting of separate image and text encoders based on vision transformers and BERT paired with a specialized instance decoder. The researchers also introduced learnable tokens called instance queries that act like cluster centroids to aggregate specific visual features from the encoded image. The model was trained on a large scale e-commerce dataset from a popular website which consisted of 15 million distinct products and over 100 million diverse images covering approximately 9000 categories including clothes, daily necessities, and instruments. Key metric results for zero shot product classification showed that the model achieved 43.8 percent and 44.8 percent accuracy on the M5Product dataset, significantly exceeding the performance of CLIP and BLIP [10]. Some limitations include background interference sensitivity, as the model still faces challenges in focusing solely on the product instance when background noise is extreme, and a heavy reliance on multi source data, meaning performance may vary if only single images are available for an entity.

2.11 RA-CLIP (Retrieval Augmented Contrastive Language-Image Pre-training) 2023 - CVPR

The primary objective of RA-CLIP was to improve efficiency in data utilization to address the data hungry nature of vanilla CLIP models, which typically require massive datasets to memorize various semantic concepts. It adopted an open book learning approach where the model utilizes a cheat sheet reference set to recognize visual concepts rather than relying solely on weights. The methodology employs a dual encoder backbone consisting of a vision transformer for images and a transformer based encoder like BERT for text, where the total image text pairs are split into a training set and a disjoint hold out reference set. For pretraining, the model primarily used a 15 million English subset of the YFCC100M dataset, along with YFCC, CC12M, and LAION, with 1.6 million pairs sampled for the reference set by default. Key metric results showed that in zero shot image classification, the model outperformed vanilla CLIP by a large margin, achieving a 12.7 percent averaged improvement across 10 datasets. Specifically, ImageNet performance reached 53.5 percent top 1 accuracy on zero shot classification compared to 37.7 percent for the vanilla CLIP baseline [11]. Some limitations include retrieval dependency, as performance is highly sensitive to the scale and quality of the reference set, as well as increased computational complexity; while retrieval can be precomputed offline for the reference set, the online retrieval process for new queries adds an additional layer of

complexity.

2.12 FLIP (Scaling Language-Image Pre-training via Masking) 2023 - CVPR

Fast Language Image Pre training (FLIP), a highly efficient method for training CLIP models that allows researchers to scale model size, dataset size and training schedules without facing prohibitive computational costs. FLIP randomly masks out a large portion of image patches around 50 to 75% during training applying the vision transformer encoder exclusively to the visible patches it optimize a contrastive loss between the encoded visible image patches and the text deliberately committing any image reconstruction loss to maximize speed, Arif and unmasked training is optionally done at the end of end of training to raise the distribution gap for inference. The models are primarily pre-trained on the LAION 400 meter data set with extensive scaling experiments utilizing the massive LAION2B Data set. FLIP achieves a 3.7x training speedup compared to standard CLIP without sacrificing accuracy, which saves thousands of TPU-days. On ImageNet-1K zero-shot transfer, a FLIP ViT-L/14 model reaches 74.6% accuracy, outperforming the equivalent OpenCLIP baseline (72.8%) [12]. However, applying masking creates a distribution shift between the training phase and full image inference making the unmasked tuning phase necessarily for optimal results while masking drastically improves speed if applied to the text site provides only marginal speed benefits because the tax encoder accounts for merely 4.4% of the total computation.

3. Comparative Table

Table 1: Comparative Table of Vision Models

Model / Method	Accuracy	Speed / Efficiency	Dataset Used
CLIP	Matches original ResNet-50 with 76.2% on ImageNet zero-shot. Outperforms fully supervised linear classifiers on 16 of 27 evaluated datasets.	Enables zero-shot transfer without dataset-specific training, but struggles with abstract tasks and out-of-distribution data.	WIT (WebImageText): 400 million image-text pairs collected from the internet.

Model / Method		Accuracy	Speed / Efficiency	Dataset Used
Vision Transformer (ViT)		Reaches 88.55% on ImageNet, 94.55% on CIFAR-100, and 77.63% on VTAB.	Requires substantially fewer computational resources to train compared to CNNs, though it lacks CNN-like inductive biases.	ImageNet (1.3M), ImageNet-21k (14M), and the in-house JFT-300M dataset (303M).
BLIP		+2.7% improvement in average recall@1 on COCO image-text retrieval; outperforms ALBEF by +1.64% on VQA.	Transfers flexibly to understanding and generation tasks, but parameter sharing causes confirmation bias.	14M images (COCO, Visual Genome, Conceptual Captions, SBU) and 115M images from LAION400M.
BLIP-2		Outperforms Flamingo80B by 8.7% on zero-shot VQAv2.	Highly efficient; utilizes 54x fewer trainable parameters than Flamingo80B by bootstrapping frozen models.	129 million images from sources like COCO, Visual Genome, CC3M, CC12M, SBU, and LAION400M.
LLaVA		Yields an 85.1% relative score vs. GPT-4 on a synthetic dataset; achieves 92.53% SOTA on Science QA when ensembled with GPT-4.	Connects the vision encoder with a language decoder using a simple linear layer, making it efficient for end-to-end fine-tuning.	595K filtered CC3M pairs + 158K language-image instruction-following samples generated using COCO images.
FLIP		74.6% accuracy on ImageNet-1K zero-shot (beating OpenCLIP’s 72.8%). Joint scaling hits 78.8% accuracy.	Achieves a 3.7x training speedup compared to standard CLIP, saving thousands of TPU-days by randomly masking patches.	LAION-400M and extensive scaling experiments utilizing LAION-2B.
InstructBLIP		Achieved SOTA on 13 held-out datasets (15% relative improvement over BLIP-2) and 90.7% on ScienceQA when fine-tuned.	Training is highly complex, requiring careful balancing strategies and significant computational resources.	26 datasets across 11 task categories (spanning text, image, and video).

Model / Method	Accuracy	Speed / Efficiency	Dataset Used
ALIGN	Outperforms BLIP-2 in conversational quality, generating rich and detailed image descriptions.	Cost-efficient; instead of training the entire model, only a small linear projection layer is trained.	Stage 1: Conceptual Captions and LAION. Stage 2: ~3.5K high-quality curated dataset.
Flamingo	Set a new few-shot SOTA on 16 multimodal benchmarks with just 32 examples.	Reduces computational complexity by mapping visual features to 64 fixed tokens, but is highly computationally intensive to run.	M3W (multimodal massive web), LTIP, and LAION.
OFA	Scored 82 on VQA and 91.2 on SNLI-VE test. CIDEr score of 154.9 for image captioning on MSCOCO.	Provides a unified, task-agnostic framework eliminating modality-specific customizations.	CC12M, SBU, Visual Genome, OpenImages, ImageNet-21K, and Object365.
ECLIP	Zero-shot product classification: 43.8% (ViT-B) and 44.8% (ViT-L) on M5Product, significantly beating CLIP and BLIP.	Scalable to a wide range of e-commerce applications like product retrieval.	15 million distinct products and over 100 million diverse images from an e-commerce website.
RA-CLIP	Zero-shot ImageNet hits 53.5% top-1 accuracy. Achieves a +12.7% averaged improvement across 10 datasets vs. vanilla CLIP.	Highly efficient data utilization using an open-book (cheat sheet) learning approach, though the online retrieval process adds complexity.	15M English subset of YFCC100M, CC12M, LAION, and 1.6M pairs sampled for a reference set.

4. Gap Analysis

While existing research in vision-language models such as CLIP, BLIP, BLIP-2, and LLaVA demonstrates strong performance in image captioning and multimodal understanding, several gaps remain when applying these methods to real-world e-commerce scenarios. Most models are trained on general-purpose datasets and generate generic captions, lacking the ability to produce domain-specific, structured, and persuasive product descriptions required in online marketplaces.

Furthermore, many approaches rely heavily on visual data and do not effectively integrate structured metadata, which is essential for accurate product representation. Issues such as

hallucinations, bias, and weak fine-grained attribute recognition are also common across models like BLIP-2 and LLaVA. In addition, there is limited focus on region-specific datasets, such as Pakistani e-commerce products, which differ in style, language, and presentation.

Our project addresses these gaps by combining image classification with an LLM-based description generator, tailored toward product-level understanding. It emphasizes structured outputs, metadata integration, and adaptability to localized product domains, making it more practical for real-world deployment.

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